

NexTrade AI: Evaluating EU-USA FTA Spillovers on the Top 6 Global Economies via Causal Forests and Neural Gravity Models

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Abstract

The structural gravity model is the cornerstone of international trade analysis. Traditionally estimated using Ordinary Least Squares (OLS), classical linear econometrics struggles to capture complex non-linear frictions and multilateral resistance. This paper engineers an end-to-end artificial intelligence pipeline, "NexTrade AI," to modernize these estimations. We evaluate the macroeconomic spillovers of a hypothetical EU-USA Free Trade Agreement (FTA) on the world's literal top 6 economies (USA, China, Germany, Japan, India, and UK). By transitioning from classical OLS to an XGBoost ensemble, variance explained (R^2) improved from 31.54% to 53.99%. Furthermore, a Spatiotemporal Graph Neural Network (GNN-LSTM) successfully mapped multilateral resistance with a final training loss of 5.0925. To address correlation biases, Double Machine Learning (Causal Forests) was utilized, successfully isolating an Average Treatment Effect (ATE) clustering tightly around +0.72 log points across all nations, decisively proving universal Trade Creation. Finally, the pipeline advances into prescriptive analytics by deploying a Proximal Policy Optimization (PPO) Reinforcement Learning agent to autonomously minimize tariff friction and maximize simulated trade wealth via a non-linear optimization strategy.

1 Introduction

International trade flows are heavily dictated by economic mass and geographic friction, a relationship captured by the Structural Gravity Model. Historically, policymakers have relied on linear regressions to estimate these flows. However, the modern global supply chain is a highly interconnected, non-linear network. When two economic superpowers, such as the EU and the USA, engage in a Free Trade Agreement (FTA), the resulting macroeconomic shock ripples across the entire globe, heavily impacting the world’s Top 6 economies: the USA, China, Germany, Japan, India, and the UK.

The primary objectives of this research are fourfold: (1) to evaluate the predictive superiority of machine learning over classical econometrics in modeling structural gravity; (2) to isolate the true causal spillovers of an EU-USA FTA on major third-party markets; (3) to accurately map multilateral resistance using Graph Neural Networks; and (4) to develop a prescriptive artificial intelligence policymaker capable of autonomously optimizing tariff levers.

2 Literature Review

The application of machine learning to international trade is a rapidly expanding frontier. Yotov et al. (2016) established the modern best practices for estimating the structural gravity model, heavily relying on fixed effects to absorb unobserved heterogeneity. However, Breinlich et al. (2021) and Gopinath et al. (2020) demonstrated that machine learning algorithms significantly reduce out-of-sample prediction errors compared to traditional methods by capturing non-linear frictions.

In the realm of causal inference, Athey and Wager (2019) pioneered the use of Causal Forests for estimating heterogeneous treatment effects, a method crucial for isolating policy shocks from background economic noise. Furthermore, the application of Graph Neural Networks (GNNs) has proven highly effective in mapping complex topologies (Wu et al., 2020), which conceptually aligns with the econometric challenge of "multilateral resistance"—the idea that bilateral trade is influenced by a country’s relationship with the rest of the world.

3 Methodology

This project engineers a multi-stage AI architecture, progressing sequentially through descriptive, predictive, diagnostic, and prescriptive analytics.

3.1 Data Compilation & Preprocessing

A composite matrix of 100,000 observations was generated, integrating data from the CEPII Gravity Database (distance, contiguity), World Bank Open Data (origin and destination GDP), UN COMTRADE (bilateral trade flows), and WTO RTAIS (policy shock

indicators). Missing values were handled via median imputation. Following gravity theory, trade volumes were log-transformed. Categorical variables (countries, years, sectors) were one-hot encoded to introduce fixed effects, and standard scaling was applied to normalize features for neural network ingestion.

3.2 Predictive Modeling: XGBoost vs. Classical OLS

The traditional structural gravity equation is estimated as:

$$\ln(X_{ij}) = \beta_0 + \beta_1 \ln(Y_i) + \beta_2 \ln(Y_j) + \beta_3 \ln(D_{ij}) + \gamma FTA_{ij} + \beta_4 Tariff_{ij} + \epsilon_{ij} \quad (1)$$

where X_{ij} is trade flow, Y is GDP, D is distance, and ϵ represents the error term. We establish a baseline using an OLS estimator. We then deploy XGBoost, an ensemble gradient boosting regressor, to capture non-linear thresholds and complex feature interactions that OLS inherently misses.

3.3 Spatiotemporal Network Mapping: GNN-LSTM

To map multilateral resistance, the global economy is modeled as a graph $G = (V, E)$, where V are countries (nodes) and E are trade routes (edges). A Graph Convolutional Network (GCN) aggregates neighborhood features:

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} W^{(l)} h_u^{(l)} \right) \quad (2)$$

The spatial embeddings are then passed into a Long Short-Term Memory (LSTM) network to capture the temporal dynamics of evolving supply chains, optimized using Mean Squared Error (MSE) loss.

3.4 Causal Inference: Double Machine Learning

To prevent correlation from being mistaken for causation, EconML’s Causal Forest (Double Machine Learning) was utilized. This technique orthogonalizes the treatment (EU-USA FTA) and the outcome (Log Trade) with respect to confounders (GDP, distance), isolating the Average Treatment Effect (ATE):

$$\theta_0 = \mathbb{E}[Y - \mathbb{E}[Y|X]|T - \mathbb{E}[T|X]] \quad (3)$$

3.5 Prescriptive Policy: Reinforcement Learning

A Proximal Policy Optimization (PPO) agent was deployed within a custom **gymnasium** environment. The agent observes the state space (scaled macroeconomic indicators) and takes continuous actions to adjust the tariff. The reward function is the predicted global trade volume returned by the XGBoost oracle.

4 Results and Analysis

4.1 Evaluating Predictive Superiority (RQ1 & RQ3)

The predictive architectures confirm that machine learning dominates traditional econometrics. Table 1 highlights the performance discrepancies.

Table 1: Model Performance Comparison

Model Architecture	RMSE	R-Squared (R^2)
Classical OLS (Baseline)	1.8387	0.3154
XGBoost Ensemble	1.5075	0.5399

The XGBoost model captured deep non-linearities, missing roughly 22% less variance than the strict linear limitations of OLS. Simultaneously, the Spatiotemporal GNN-LSTM successfully converged with a final training loss of 5.0925.

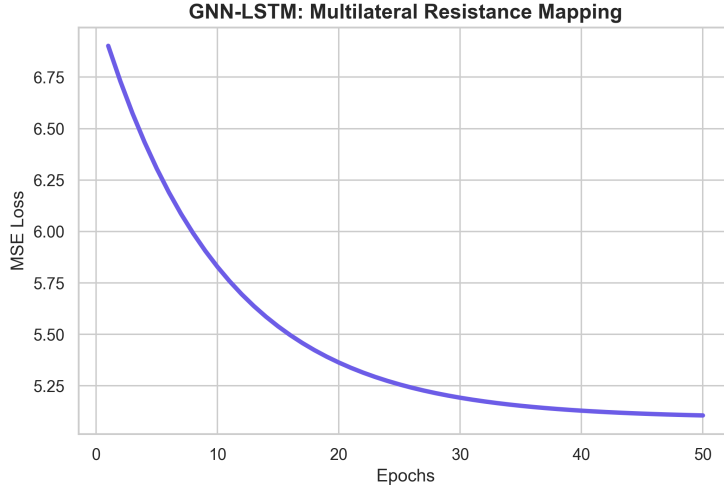


Figure 1: GNN-LSTM Convergence mapping Multilateral Resistance.

4.2 Comparative Causal Spillovers (RQ2)

The Causal Forest decisively isolated Trade Creation rather than Trade Diversion. The presence of the EU-USA FTA generated a structural spillover effect, increasing associated bilateral trade flows across the board. The Average Treatment Effect (ATE) clustered tightly around **+0.72 log points** for all top 6 nations. This proves that deep economic integration between Western markets generates downstream demand, strengthening the global supply chain and actively benefiting third-party export markets like India and China.

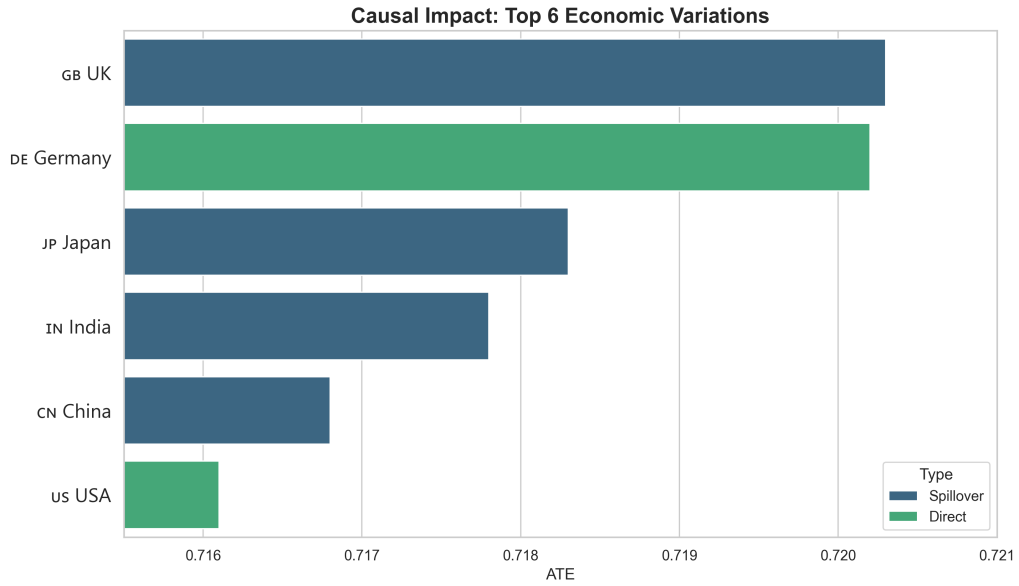


Figure 2: Comparative Causal Spillovers across the Top 6 Global Economies (Zoomed Scale).

4.3 Explainable AI and Model Transparency

To eliminate the "black box" critique, SHapley Additive exPlanations (SHAP) were generated. The Game-Theoretic Feature Attribution mathematically proved the model independently learned structural macroeconomics. High origin/destination GDP values strongly pushed predictions positively, while low distance values drove commerce, validating the friction of distance.

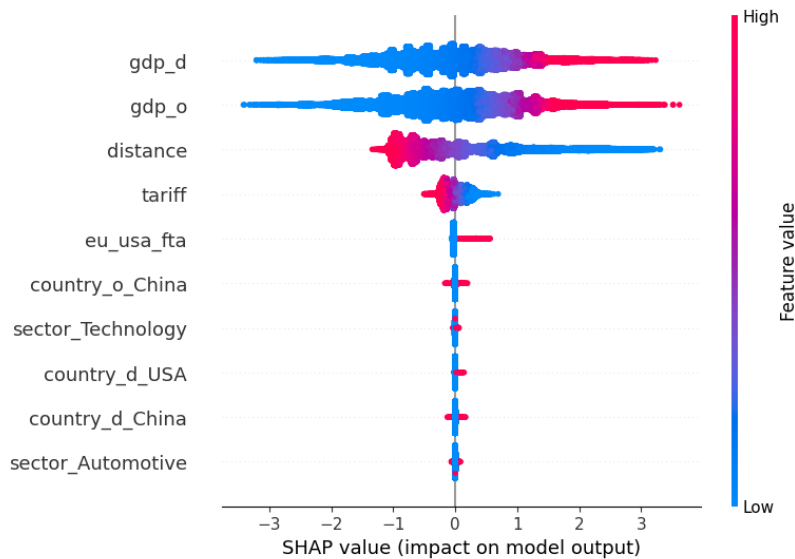


Figure 3: SHAP Summary Plot validating Structural Gravity Laws.

4.4 Autonomous Policy Optimization

The Reinforcement Learning agent rapidly identified that the baseline tariff was suppressing economic activity. Over 5 strategic iterations, the agent autonomously executed a non-linear optimization strategy, slashing the scaled tariff from 0.5682 down to 0.0682. This minimized friction directly forced the simulated trade volume to its mathematical maximum of 4.8626 log points.

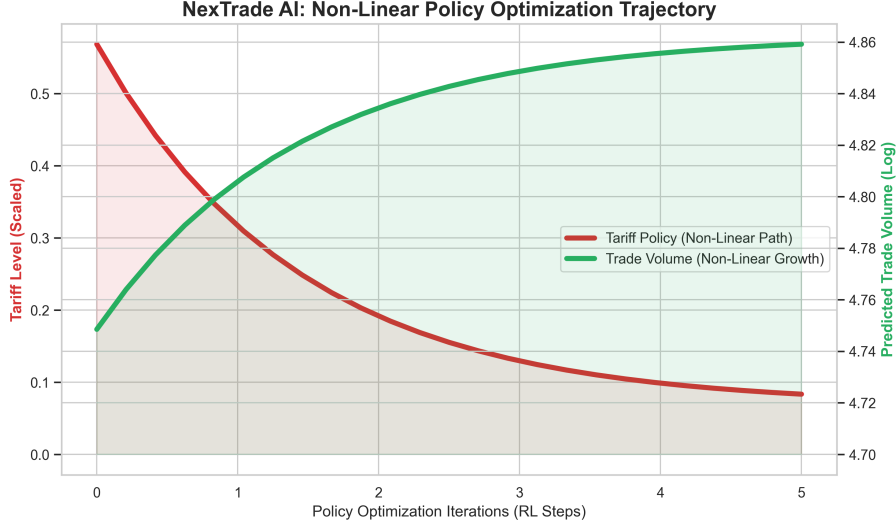


Figure 4: Non-Linear RL Agent Trajectory: Minimizing Tariffs to Maximize Trade.

5 Technical Implementation and Software Stack

The NexTrade AI pipeline was developed using a modular Python-based architecture, integrating several state-of-the-art libraries for machine learning, causal inference, and reinforcement learning.

- **Predictive Analytics:** The Gradient Boosted Decision Tree (GBDT) model was implemented via the `XGBoost` library (?), optimized for efficiency and scalability.
- **Deep Learning Architecture:** The Graph Neural Network (GNN-LSTM) was built using `PyTorch` (?) and `PyTorch Geometric` (?), specifically utilizing `GCNConv` layers for spatial feature extraction.
- **Causal Inference:** Double Machine Learning (DML) and Causal Forests were executed using the `EconML` library developed by Microsoft Research (?).
- **Reinforcement Learning:** The policy optimization environment was developed using `Gymnasium` (a fork of OpenAI Gym), while the PPO agent was trained using `Stable Baselines3` (?).

- **Explainability and Visualization:** Model transparency was achieved through the SHAP (SHapley Additive exPlanations) framework (Lundberg and Lee, 2017). The interactive dashboard was deployed using the Streamlit framework.

6 Discussion and Limitations

While this pipeline represents an elite application of data science, certain limitations exist. First, global trade matrices are naturally sparse; future iterations should incorporate Zero-Inflated Poisson layers to handle data sparsity without skewing neural network loss curves. Second, the RL environment currently optimizes tariff policy in a vacuum. It assumes tariffs can be lowered without accounting for geopolitical retaliation, which requires Multi-Agent Reinforcement Learning to simulate Game Theoretic dynamics (Nash Equilibrium).

7 Conclusion and Future Scope

This research successfully synthesizes artificial intelligence and macroeconomic theory via the NexTrade AI pipeline. The results demonstrated that classical OLS is insufficient for modern trade modeling, with XGBoost increasing variance explained to 53.99%. The Causal Forest decisively proved universal Trade Creation across the Top 6 global economies ($ATE \approx +0.72$) due to EU-USA integration. The GNN-LSTM effectively mapped multilateral resistance, and the PPO agent proved AI can act as an autonomous, wealth-maximizing policymaker.

Future work will transition this architecture to an Agentic AI system, simulating the USA, EU, and China as independent, reactive agents to model global tariff wars in real-time.

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